

2023 AMC 10B Problem 18 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 18. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 18

**Problem:**

Suppose that a , b , and c are positive integers such that

$$\frac{a}{14} + \frac{b}{15} = \frac{c}{210}$$

Which of the following statements are necessarily true?

- I.* If $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both, then $\gcd(c, 210) = 1$.
- II.* If $\gcd(c, 210) = 1$, then $\gcd(a, 14) = 1$ or $\gcd(b, 15) = 1$ or both.
- III.* $\gcd(c, 210) = 1$ if and only if $\gcd(a, 14) = 1$ and $\gcd(b, 15) = 1$.

Answer Choices:

- A. *I*, *II*, and *III*
- B. *I* only
- C. *I* and *II* only
- D. *III* only
- E. *II* and *III* only

Solution:

Note that 14 and 15 are relatively prime and $210 = 14 \cdot 15$. Let q and r be two relatively prime positive integers (such as 14 and 15). The equation $\frac{a}{q} + \frac{b}{r} = \frac{c}{qr}$ is equivalent to $ar + bq = c$. Hence

$$\gcd(c, q) = \gcd(ar + bq, q) = \gcd(ar, q) = \gcd(a, q)$$

Similarly, $\gcd(c, r) = \gcd(b, r)$. Also, $\gcd(c, qr) = 1$ if and only if $\gcd(c, q) = 1$ and $\gcd(c, r) = 1$. Therefore Statement III is true, and that statement logically implies Statement II. To see that Statement I might be false, let $a = 1$, $b = 3$, and $c = 57$; then the hypothesis is true but the conclusion is false. Thus Statements (E) *II and III* are the only ones that are necessarily true.

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